

SUMAN LAMSAL

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Summary

ML engineer with three years of backend and data engineering experience, finishing an M.S. in Computer Science at Indiana University. Recent work includes a pose estimation pipeline built for a neuroscience lab, an unlearning-based setup for testing whether LLM reasoning is causal, and a published model for heart rate monitoring on wearables. Day to day I work in PyTorch, Hugging Face, Docker, and on Slurm GPU clusters. Looking for full-time ML engineering roles.

Education

Master of Science in Computer Science

Indiana University Indianapolis

Aug 2024 – Dec 2026

Indianapolis, IN | GPA: 3.78 / 4.0

Bachelor of Engineering in Electronics, Communication & Information Engineering

Tribhuvan University, Institute of Engineering (Pulchowk Campus)

Nov 2019 – May 2023

Lalitpur, Nepal | GPA: 3.88 / 4.0

Technical Skills

Languages: Python, SQL, Java, C/C++, Bash, R

ML: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, DeepLabCut, OpenCV, LangChain

Infrastructure: Docker, Slurm, Git, GitHub Actions, MLflow, Airflow, Linux, AWS (S3, EC2)

Data: NumPy, Pandas, SciPy, PostgreSQL, MySQL, Matplotlib, Seaborn, Tableau, Jupyter

Professional Experience

Backend Engineer

OCM Engineering Groups, LLC

Aug 2023 – Aug 2024

Greenwood, IN

- Built data ingestion and validation pipelines for utility inspection platforms, cutting manual data prep time by 20% across three client deployments.
- Automated GIS record updates and pole-inventory workflows, which reduced data-entry errors for telecom and utility clients.
- Maintained backend APIs for data retrieval and reporting on inspection systems that field crews depended on daily.

AI / ML Fellow

Fusemachines

Jan 2023 – Jun 2023

Kathmandu, Nepal

- Trained and tuned logistic regression, random forest, and CNN models for predictive analytics tasks, improving accuracy by 15% through feature engineering and hyperparameter search.
- Built text classification and image recognition prototypes in PyTorch and TensorFlow.
- Set up reproducible experiment pipelines with Scikit-learn, Pandas, and MLflow so runs could be repeated and audited.

Data Engineering Intern

Everestwalks Groups

Jun 2022 – Nov 2022

Kathmandu, Nepal

- Rebuilt ETL pipelines in Python, SQL, and Pandas, cutting data latency by 25%.
- Automated cleaning and transformation jobs with Bash and Airflow, removing manual steps from four recurring workflows.

Research Experience

Independent Research, Computer Vision for Gait Analysis

Indiana University Indianapolis (FADS Program)

Jan 2026 – May 2026

Indianapolis, IN

- Built an automated gait analysis pipeline for a neuroscience group studying Down syndrome mouse models, replacing a manual video review process that took hours per recording.
- Trained a ResNet-50 pose estimation model in DeepLabCut on 194 labeled frames to track six anatomical landmarks. Ran 100,000 iterations to a final training loss of 0.0023 and test error of 8.75 px.
- Moved training onto IU's Quartz cluster (V100, CUDA 11.8, Slurm) and wrote a submission tool that handles project transfer, job scripts, and Windows-to-Linux path conversion. Cut training wall-clock time by over 95% compared to running locally.
- Added a quality-control layer with likelihood filtering, interpolation, and temporal smoothing to stabilize noisy pose estimates, plus a calibration utility and a batch parameter extractor. All three tools are documented for reuse.
- Checked every output against published biological ranges before trusting it. Velocity came out at 11.34 cm/s, hindpaw stride at 3.04 cm, and bilateral asymmetry at 1.0%, all within expected bounds.

Selected Projects

Chain-of-Thought Faithfulness in Domain-Adapted LLMs | *PyTorch, Hugging Face* 2026

- Built the machine unlearning pipeline (NPO with KL regularization) used to test whether a model’s chain-of-thought reasoning actually drives its predictions. Handled MLP layer selection, checkpointing, and integration with the evaluation code.
- Extended parametric faithfulness measurement from Tutek et al. (EMNLP 2025) to BioMistral-7B, Saul-7B, and Mistral-7B-Instruct across 412 reasoning steps on MedQA and LegalBench, running on A100-80GB nodes.
- Found that domain-adapted reasoning is just as faithful as the base model (FF-HARD around 67%, $p=0.96$ after controlling for unlearning efficacy) but much harder to erase: 70% of unlearning attempts failed versus 21% for Mistral. We called this reasoning entrenchment.
- Traced a catastrophic forgetting failure to hyperparameters that worked on medical text but collapsed the model on legal text, dropping specificity from 61% to 16% and invalidating that run.

FairCFRNet: Path-Specific Causal Fairness for Treatment Effects | *PyTorch, Causal Inference* 2026

- Sole-authored a method that adds differentiable path-specific fairness constraints to CFRNet. Existing methods balance treatment group distributions but can’t block a specific unfair causal pathway, which is the gap this closes.
- Wrote a Differentiable Causal Estimation module that computes path-specific effects inside the network’s computational graph using AIPW estimation and counterfactual mediators, so the whole thing stays end-to-end trainable.
- On IHDP, cut the demographic parity gap by 22% (0.695 from 0.891) and improved PEHE by 6.1% (0.844 from 0.899) at the same time. ITE correlation rose from 0.367 to 0.445.
- Ran ablations across regularization strengths and found the model collapses at lambda 2.0 (correlation drops to -0.088), which sets the usable range at roughly 0.3 to 0.7.

Transformer vs. Mamba Benchmark | *NLP, PyTorch, Hugging Face* 2024

- Compared BERT-base (110M) against Mamba-370M on AI-generated text detection using a balanced 25K-sample subset of a 347K corpus. Both hit F1 of 0.99 (BERT 0.9954, Mamba 0.9922).
- Measured inference cost across input lengths. Mamba stayed flat at about 0.06s from 500 to 5,000 words while BERT climbed to roughly 1.1s, so about 18x faster at long inputs despite having 3.4x more parameters.
- Wrote the fine-tuning pipelines with custom classification heads and class-balanced sampling that preserved the original text length distribution. Evaluated on an A100 with confusion matrices, F1, and AUROC.

Publications

N. Bastakoti, S. Poudel, **S. Lamsal**, Y. Sharma, A. Jaiswal. “Heart Rate Estimation from PPG and Accelerometer via CNN-Transformer Fusion: HeartFusion.” *International Conference on Technologies for Computer, Electrical, Electronics & Communication (ICT-CEEL)*, pp. 94–96, 2023.

- CNN and transformer model that predicts heart rate as a distribution rather than a point estimate, so it reports its own confidence. MAE of 4.82 bpm on IEEE SPC 2015 and 6.20 bpm on PPG-DaLiA under leave-one-session-out cross-validation, using 151K parameters against Deep PPG’s 8.5M.